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SUPPLEMENTARY MATERIAL

The Gram Matrix of the Tetrad Equation

Stephen Nelson

Field Dynamics Research

E-mail: stephen@fielddynamics.org

Abstract
The tetrad equation $g = e^T \eta\, e$ is even in the spatial frame, so its spatial realisation M comes with a parity partner $-M$. We study the Gram matrices of the pair. Self-coupling is root-blind and cross-coupling reverses every sign; since the cross-coupling is also unchanged by exchanging the roots, Gram data resolve whether two factors come from the same root or from opposite roots, never which root is which. For the tetrahedral frame, bordering M with a row of ones gives a Hadamard matrix, which yields $M^T M = 4I - J$, the spectral split $I = \frac{1}{4}G + \frac{1}{4}J$ and the rows of M as eigenvectors, at once. The Gram matrix fixes M up to a unique element of $O(3)$, and that element is $-I$ for the two roots: the roots are told apart by orientation alone, which flips because three is odd. The two roots are the parity classes $xyz = \pm 1$ of the cube, and the symmetry group of their compound, the stellated octahedron, is that of one tetrahedron times $\{\pm I\}$. On the connection, exchange of the lower indices splits the 64 components into 40 and 24; the torsion-free condition is the projection onto the 40, and for a metric connection the discarded 24 are exactly the torsion, the only place a connection can tell the roots apart. Every step is exact and machine-checked.

Keywords: tetrad, Gram matrix, spatial parity, Hadamard matrix, stellated octahedron, torsion, proof

Setting

A Lorentz tetrad e satisfies the tetrad equation

$$g = e^T \eta\, e, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1). \quad (1)$$

Spatial parity $P = \mathrm{diag}(1, -1, -1, -1)$ acts on the frame index, $e \mapsto Pe$. Since $P^T \eta P = \eta$, the metric is unchanged, $(Pe)^T \eta (Pe) = g$, while $\det(Pe) = -\det e$: the equation has two roots, carried into each other by P, and the metric sees only $(\det e)^2$ [3].

The four legs are realised in space as in [4]: a *spatial realisation* is a matrix $M \in \mathbb{R}^{3\times 4}$ whose column v_a represents leg a, with every pair of legs in the same relation and every leg of the same magnitude. Those premises force the Gram matrix and the regular tetrahedron, unique up to $O(3)$. We take that result as given, normalise the magnitude to $|v_a|^2 = 3$, and use the representative

$$M_1 = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix}. \quad (2)$$

Parity reflects the realisation, $v_a \mapsto -v_a$, so the second root is realised by $M_2 = -M_1$. We write I for the identity, J for the 4×4 all-ones matrix and $\mathbf{1} = (1, 1, 1, 1)^T$.

Part I: The three Gram products

Definition 1. For realisations U, W the Gram product is $G_{UW} = U^T W$, with entries $v_a \cdot w_b$. From the parity pair we form $G_{M_1 M_1}$, $G_{M_2 M_2}$, $G_{M_1 M_2}$ and $G_{M_2 M_1} = G_{M_1 M_2}^T$. We write $G = G_{M_1 M_1}$.

Lemma 1 (Root-blindness)**.** $G_{M_2 M_2} = G_{M_1 M_1}$.

Proof. $(-M_1)^T(-M_1) = M_1^T M_1$. □

Lemma 2 (Cross-coupling)**.** $G_{M_1 M_2} = G_{M_2 M_1} = -G$. *Hence the Gram products take the value G on a same-root pair and $-G$ on a cross-root pair, and each value is unchanged when the two roots are exchanged.*

Proof. $M_1^T(-M_1) = -G = (-M_1)^T M_1$. Exchanging the roots sends $G_{M_1 M_1} \leftrightarrow G_{M_2 M_2}$ and $G_{M_1 M_2} \leftrightarrow G_{M_2 M_1}$, which are equal pairs. □

So the cross-coupling is *root-resolving* in a precise sense: it resolves the *relative* root of two factors. No Gram product says which root a single frame is; Proposition 1 gives the quantity that does.

Lemma 3 (Explicit Gram matrix)**.** *Let $\hat{H} = \begin{pmatrix} \mathbf{1}^T \\ M_1 \end{pmatrix}$. Then $\hat{H}\hat{H}^T = 4I$, hence $\hat{H}^T\hat{H} = 4I$, $M_1\mathbf{1} = 0$ and*

$$G = M_1^T M_1 = 4I - J, \qquad M_1 M_1^T = 4I_3, \quad (3)$$

Proof. The rows of M_1 sum to zero and are mutually orthogonal with squared norm 4, by inspection of (2); with the row $\mathbf{1}^T$, of squared norm 4, this is $\hat{H}\hat{H}^T = 4I$. A square matrix with $\hat{H}\hat{H}^T = 4I$ has $\hat{H}^{-1} = \frac{1}{4}\hat{H}^T$, so $\hat{H}^T\hat{H} = 4I$. Its left side is $\mathbf{1}\mathbf{1}^T + M_1^T M_1 = J + G$. □

$\hat{H}$ is the 4×4 Hadamard matrix [5]. Entrywise, G has diagonal 3 and off-diagonal -1; with Lemmas 1 and 2 all three products are determined:

$$G_{M_1M_1} = G_{M_2M_2} = 4I - J, \qquad G_{M_1M_2} = J - 4I. \tag{4}$$

Lemma 4 (Eigenstructure)**.** *$\frac{1}{4}G$ and $\frac{1}{4}J$ are complementary orthogonal projections, $I = \frac{1}{4}G + \frac{1}{4}J$. Thus G has eigenvalue 0 on $\mathbf{1}$ (closure) and eigenvalue 4 with multiplicity 3 on $\mathbf{1}^\perp$, which is spanned by the rows of M_1: $GM_1^T = 4M_1^T$. The cross-coupling has eigenvalues 0 and -4 on the same spaces.*

Proof. $J^2 = 4J$ gives $(\frac{1}{4}J)^2 = \frac{1}{4}J$, and $\frac{1}{4}G = I - \frac{1}{4}J$ is the complementary projection. $GM_1^T = M_1^T(M_1M_1^T) = 4M_1^T$, and the three rows are independent by $M_1M_1^T = 4I_3$. □

Lemma 5 (Invariance and rigidity)**.** *For $Q \in O(3)$ the three products are unchanged by $(M_1, M_2) \mapsto (QM_1, QM_2)$. Conversely, if W has $W^TW = G$, then $W = QM_1$ for a unique $Q \in O(3)$, namely $Q = \frac{1}{4}WM_1^T$. For $W = M_2$, $Q = -I$.*

Proof. $(QU)^T(QW) = U^TQ^TQW = U^TW$. The converse is the rigidity lemma of [4], since G has rank 3; multiplying $W = QM_1$ on the right by M_1^T and using $M_1M_1^T = 4I_3$ gives Q. For $W = -M_1$ this is $-I$. □

So the Gram matrix is the complete invariant of a realisation under $O(3)$, and the two roots are one $O(3)$ orbit: they are related by $-I$, which has determinant -1 and lies outside $SO(3)$.

Proposition 1 (Orientation)**.** *For each triple of legs $a < b < c$, the orientation $\omega_{abc}(M) = \det(v_a, v_b, v_c)$ is nonzero on M_1, invariant under $SO(3)$, and satisfies $\omega_{abc}(M_2) = -\omega_{abc}(M_1)$.*

Proof. Any three columns of M_1 are independent, since $\mathbf{1}$ spans the null space of M_1. $\det(Qv_a, Qv_b, Qv_c) = \det Q\, \omega_{abc}$, and $\det(-I_3) = (-1)^3 = -1$. □

This is the spatial counterpart of $\det(Pe) = -\det e$: the roots differ only in orientation, and orientation flips because $D = 3$ is odd.

Part II: Geometric identification

Theorem 1 (Tetrahedra in the cube)**.** *The columns of M_1 are the four corners of the cube $\{\pm 1\}^3$ with $xyz = +1$, and those of M_2 the four with $xyz = -1$. Each set is a regular tetrahedron centred at the origin, with every edge of squared length 8. The two sets are disjoint and together are the eight corners of the cube; their compound is the stellated octahedron $S = [\, M_1 \mid M_2 \,]$ [11, 12].*

Proof. Each column of (2) has an even number of entries -1, so $xyz = +1$; negating three coordinates multiplies xyz by $(-1)^3$. The corners with $xyz = +1$ are four in number, so they are exactly the columns of M_1, and the rest are those of M_2. By Lemma 3, $|v_a - v_b|^2 = 3 + 3 - 2(-1) = 8$ for $a \neq b$, and $\sum_a v_a = M_1\mathbf{1} = 0$. □

Following [1, 3], the first tetrahedron is TetA (the identity component under spatial parity) and the second TetB (the parity component).

Theorem 2 (Symmetry and non-isomorphism)**.** *The orthogonal maps permuting the corners of TetA form a group of order 24, acting on the four legs as the whole of S_4. Those permuting the corners of S form the group*

$$\mathrm{Sym}(S) = \mathrm{Sym}(\mathrm{TetA}) \times \{I, -I\}, \tag{5}$$

of order 48, and every element outside Sym(TetA) *carries TetA onto TetB. In particular TetA is not congruent to S.*

Proof. A map in Sym(TetA) is fixed by the permutation it induces, as the corners span $\mathbb{R}^3$. Every permutation occurs: by Lemma 5, $M_1\Pi$ has Gram matrix $\Pi^TG\Pi = G$ for any permutation matrix Π, so $M_1\Pi = QM_1$ for some $Q \in O(3)$. Hence the order is 24. Two corners of the cube are at squared distance 8 exactly when they differ in two coordinates, that is, when they have the same xyz. A map $L \in \mathrm{Sym}(S)$ is an isometry, so it carries TetA, four corners at mutual squared distance 8, onto four such corners, all with one value of xyz: onto TetA or onto TetB, and then TetB $= -$TetA onto the other. Those of the first kind form Sym(TetA). If L is of the second kind, $-L$ is of the first, since $-I$ exchanges TetA and TetB; and $-I$ commutes with every map, which gives (5). The verification script confirms the count exhaustively over all 8! relabellings of the corners. □

The extra factor is exactly parity: the compound has twice the symmetry of either root, and the doubling is the element that exchanges them.

Part III: The index grid

The affine connection $\Gamma^\lambda{}_{\mu\nu}$ has $K^3 = 64$ components. Fixing λ gives a 4×4 *slice* in (μ, ν), with 4 diagonal entries, 6 in the strict upper triangle and 6 in the strict lower triangle. Over the four slices,

$$\underbrace{4 \times 6}_{24} + \underbrace{4 \times 4}_{16} + \underbrace{4 \times 6}_{24} = 64. \tag{6}$$

In the framework's reading of the grid the upper triangle is TetA, the lower triangle TetB and the

diagonal shared, and exchanging the triangles is the TetB step [2, 7].

Lemma 6 (The exchange split)**.** *Let τ exchange the lower indices, $(\tau\Gamma)^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\nu\mu}$. Then $\tau^2 = 1$, τ fixes the diagonal and exchanges the two triangles, and $\Pi_+ = \frac{1}{2}(1+\tau)$ is a projection of rank 40 whose kernel, the τ-odd part, has dimension 24 and is the torsion $T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu}$.*

Proof. On a mirror pair with upper entry u and lower entry l, τ swaps u and l; the even combination $\frac{1}{2}(u+l)$ is kept and the odd one $u-l$, the torsion, is removed. Each slice has $4+6$ even and 6 odd directions; times four slices, 40 and 24. □

Written in the coordinates of (6), the torsion-free condition is $l = u$: it keeps the diagonal and one triangle and makes the other a copy, the form in which it reads as keeping shared and TetA and discarding TetB. Independently of which triangle is written down, what is kept is the exchange-even combination $\frac{1}{2}(u+l)$, which does not change when the triangles are exchanged, and what is discarded is the exchange-odd difference $u-l$.

Part IV: The torsion-free projection

Theorem 3 (Torsion-free projection)**.** *(i) The torsion-free condition is the projection Π_+ of the 64 components onto 40, with the 24 torsion components as kernel. It is not injective, so it is not an isomorphism:*

$$\boxed{16 + 24 = 40 \;\neq\; 64 = 24 + 16 + 24.} \qquad (7)$$

(ii) For any metric g, a connection compatible with g is determined by g and its torsion: the differences of two g-compatible connections form a 24-dimensional space, mapped bijectively onto the 24-dimensional space of torsions.

(iii) Hence, if each root is given a g-compatible connection, the two connections differ only through their torsion. A torsion-free metric connection is the Levi-Civita connection of g, which by (1) is the same on both roots.

Proof. (i) is Lemma 6. (ii) The difference Δ of two g-compatible connections satisfies $\Delta_{\lambda\mu\nu} + \Delta_{\nu\mu\lambda} = 0$, with the first index lowered by g: antisymmetry in two of three indices, $4 \times 6 = 24$ dimensions. If its torsion vanishes, Δ is also symmetric in the last two, and a tensor symmetric in one pair of slots and antisymmetric in another vanishes; so $\Delta \mapsto T(\Delta)$ is injective between spaces of equal dimension, and bijective. Its inverse is the contorsion [10]. (iii) The metric is the same on both roots (Setting), so both connections are the Levi-Civita connection of g plus the contorsion of their torsion, which vanishes in the torsion-free case. The verification script checks (ii) exactly for η and for a generic Lorentzian metric. □

Remark. Parts I and IV say the same thing at two levels. The metric, and every Gram product, cannot tell the roots apart; neither can any torsion-free metric connection, since it is built from the metric. At the connection level the only room for the distinction is the 24 torsion components that general relativity sets to zero [8] and Einstein–Cartan theory keeps [9, 10]. GFD reads the torsion as where TetA and TetB are told apart [2, 6]. The theorem proves that the torsion is the only place available; it does not prove that the torsion is nonzero, nor how it transforms under parity.

Summary

Result	Proves	Requires
Lemma 1	$G_{M_2M_2} = G_{M_1M_1}$	$M_2 = -M_1$
Lemma 2	$G_{M_1M_2} = G_{M_2M_1} = -G$	idem
Lemma 3	$G = 4I - J$	Hadamard
Lemma 4	$I = \frac{1}{4}G + \frac{1}{4}J$	Lemma 3
Lemma 5	Invariance, rigidity	[4]
Prop. 1	Orientation flips	$D = 3$ odd
Thm. 1	$xyz = \pm 1$; cube	Lemma 3
Thm. 2	Symmetry $\times\{\pm I\}$	Lemma 5
Lemma 6	$64 = 40 + 24$ under τ	Indices
Thm. 3	Roots differ only by T	Lemma 6

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